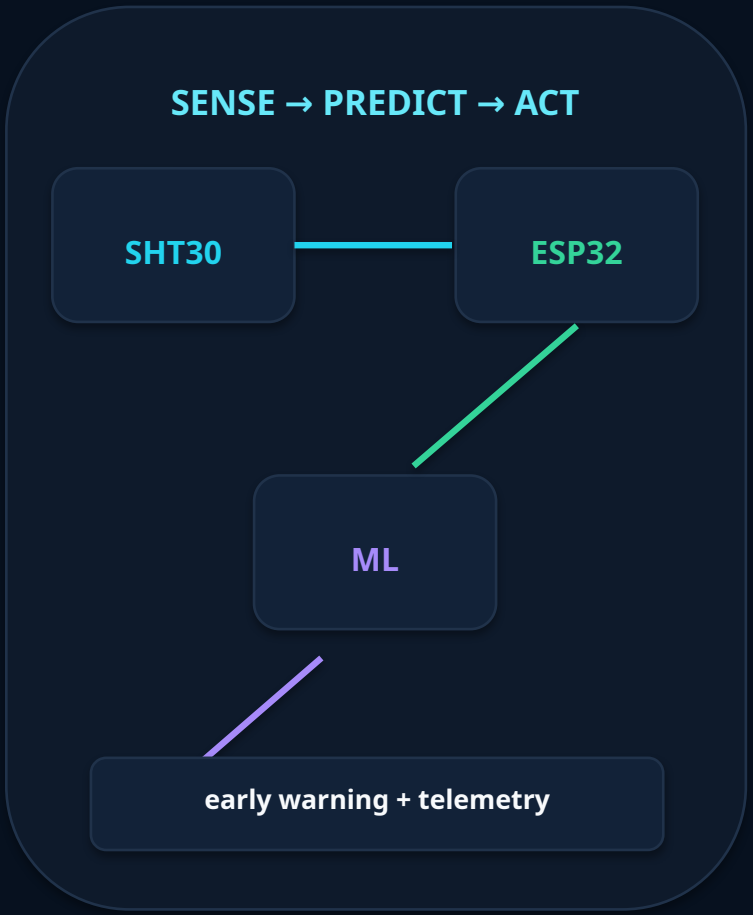
SMART INDIA HACKATHON 2025

EDGE-AI
PREDICTIVE COLD CHAIN MONITOR

FreeRTOS-enabled hyper-local monitoring with on-device predictive ML

ESP32 | SHT30 | Decision Tree | FreeRTOS | ThingSpeak HTTPS



SIH26215 · Theme: MedTech · Hardware · Team 14

IMPACT INNOVATORS · RV UNIVERSITY

Cold-chain failures are often detected after the damage starts

The project shifts monitoring from reactive thresholds to local prediction and resilient telemetry.

THE PROBLEM

Temperature-sensitive healthcare products can leave the 2–8°C band during last-mile movement.

- A threshold alarm tells the operator that a breach is happening — not how soon it will happen.
- Connectivity interruptions can break continuous remote visibility.
- Irregular sampling cadence can make a trained time-series model misleading if deployed naively.
- A low-cost field device needs local computation, buffering and recovery — not cloud-only intelligence.

DESIGN SHIFT

Reactive monitoring → **Predictive edge monitoring**

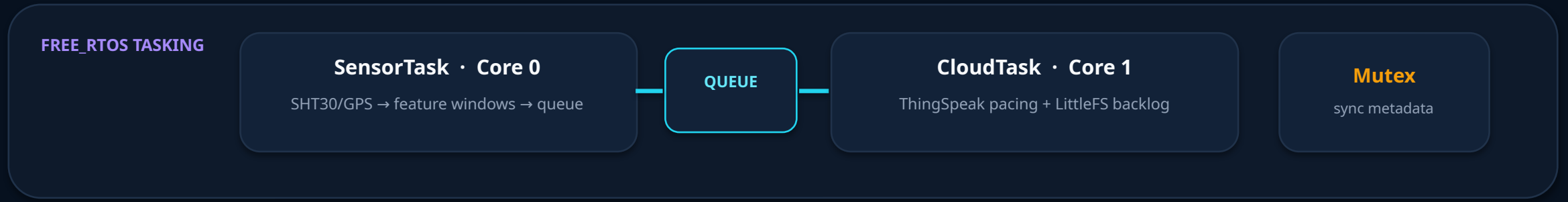
SENSE	2-second local sensing
ANALYZE	rules + thermal models
PREDICT	Decision Tree on ESP32
BUFFER	LittleFS during outages
SYNC	ThingSpeak REST/HTTPS

One device, two cadences, one resilient data path

Fast safety logic runs continuously; the trained ML model runs on the cadence it was trained for.



IMPORTANT: 2 s sensing ≠ 10 min ML inference



Built from low-cost, commonly available modules

The MVP uses a practical sensor + MCU + cloud stack while keeping intelligence at the edge.

ESP32
MCU + Wi-Fi
FreeRTOS runtime

SHT30
Temp + humidity
I^2C / 2 s reads

NEO-6M
GPS context
UART / freshness gate

LittleFS
Persistent offline
telemetry backlog

WIRING

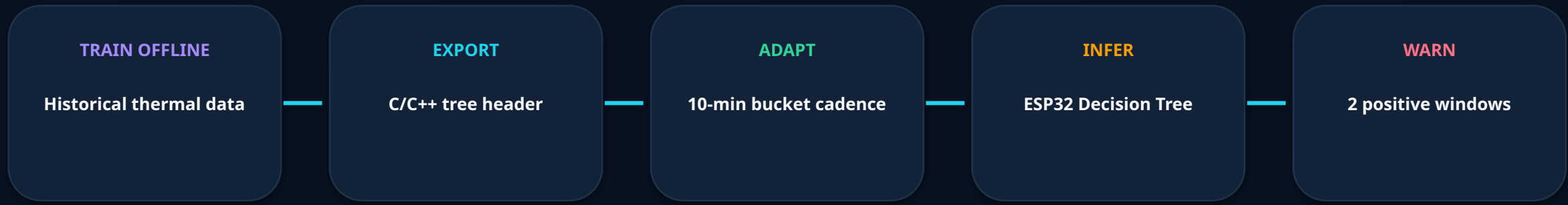
SDA	GPS RX
GPIO 21	GPIO 16
SCL	GPS TX
GPIO 22	GPIO 17

FIRMWARE STACK

Sensor task	sampling + feature window
Decision layer	heat/freeze latch + thermal models
ML layer	embedded Decision Tree
Cloud task	15 s ThingSpeak + offline backlog

The learned model runs on the ESP32 — not in the cloud

The cloud is telemetry. The trained Decision Tree is embedded into firmware for local inference.



10 FEATURES

- temperature
- 5-sample std
- 10-min delta
- rising fraction
- count above 6°C
- 5-sample mean
- 5-sample slope
- 20-min delta
- distance to 8°C
- slope acceleration

MODEL CONTRACT

Threshold	0.85
Persistence	2 windows
Domain	current bucket < 8°C

Minimum bucket coverage: 80% · five valid buckets before first inference

ML is only one layer — the system combines learned, deterministic and thermal reasoning

Each layer answers a different operational question.

FAST RULES

I

Is the product already outside the safe band?

- 2 s sampling
- heat/freeze latch
- persistence filter

EDGE ML

I

Is the trajectory likely to cross 8°C soon?

- 10 min windows
- Decision Tree
- local predictive alert

THERMAL MODELS

H

How quickly is the condition changing / accumulating exposure?

- Newton cooling
- Arrhenius exposure
- risk score

Operational principle: keep critical detection local so connectivity is not on the critical path for prediction.

Primary frozen holdout: reproducible evidence, not field-accuracy marketing

Historical Nepal thermal data were split by source file so entire files were held out from model development.

66.3%
Precision
How often a positive warning was correct

68.1%
Recall
Share of target positives detected

67.2%
F1
Balance of precision and recall

0.907
ROC-AUC
Ranking quality across thresholds

HOLDOUT
6,110
rows
12 source files

FROZEN CONFUSION MATRIX

	Predicted −	Predicted +
Actual +	175 TP	82 FN
Actual −	89 FP	5,764 TN

Threshold: 0.85

WHAT THE NUMBERS MEAN

- Validation source: historical Nepal thermal data.
- Source-file holdout: 48 development files + 12 completely held-out files.
- These metrics do not represent Bengaluru field accuracy or the final physical ThingSpeak/ML runtime.

Designed for real field constraints: connectivity loss, cadence gaps and secure telemetry

The final software release hardens the path between sensing and cloud storage.

Offline resilience

LittleFS backlog
rate-limited replay

Secure cloud path

HTTPS +
DigiCert root CA

Cadence quality

80% coverage
5 valid buckets

False-alarm control

2-window
persistence

Sensor recovery

re-init after
5 failures

GPS freshness

120 s age
gate

Important limitation: offline buffering protects telemetry data, but not real-time remote alerting while the network is disconnected.

What is verified, what is software-ready, and what still needs the physical bench

This slide is intentionally explicit about evidence boundaries.

ESP32 + SHT30 sensing	VERIFIED	Hardware
Wi-Fi path	VERIFIED	Hardware
Final secure v8.3.2 firmware	READY	Static audit
ThingSpeak physical publish	PENDING	Bench test
Physical ML runtime in 2–8°C	PENDING	~50–60 min
GPS satellite fix	DEFERRED	Time-boxed

NOT CLAIMED Bengaluru field accuracy · battery-life measurement · remote alerts during disconnection

NEXT BENCH CHECK

1. Create ThingSpeak channel + enter credentials
2. Upload v8.3.2 and verify a positive entry ID
3. Run in 2–8°C for ~50–60 min to observe physical ML inference
4. Capture ml_data_quality, probability, prediction + model version

v8.3.1 hardening: first-bucket guard · post-attempt cloud pacing · stream-safe LittleFS backlog

From validated software to a field-ready cold-chain node

The next step is not more complexity — it is physical validation in a controlled 2–8°C environment.

IMPACT

Healthcare	Earlier intervention before a breach; reduced exposure risk.
Operations	Central telemetry + local prediction + traceable backlog.
Economics	Low-cost edge hardware designed for scalable fleet use.

ROADMAP

1	**Software/model holdout**	COMPLETE
2	**ESP32 + SHT30 integration**	COMPLETE
3	**Secure ThingSpeak build**	READY
4	**Physical ML runtime 2–8°C**	NEXT
5	**Field thermal validation + GPS**	LATER

SENSE → ANALYZE → PREDICT → BUFFER → SYNC

EDGE-AI COLD CHAIN MONITOR

Sources: final v8.3.2 firmware · frozen ML holdout evidence · SIH execution status · ThingSpeak secure setup